



Hosiery Knitting

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Background

1.1 Context

This document presents the modelling assumptions and data sources for **hosiery knitting**, representing the production of socks or similar small-diameter tubular knitted goods.

The model follows the same methodological principles applied in flat and circular knitting, ensuring coherence across all knitting process datasets.

Background data: ecoinvent 3.12, system model: cut-off by classification. Background processes are linked via generic market activities (GLO/RoW).

Note on sources: machine performance data are drawn from manufacturer catalogues (Lonati/Santoni) and academic sources. Some supporting references (K&G Garment, Accio) are commercial or grey literature and are used only for indicative benchmarking of productivity and material utilisation.

1.2 Scope and functional unit

- **Functional unit:** 1 kg of hosiery fabric (average of mixed cotton and synthetic yarns).
- **System boundary:** Gate-to-gate hosiery knitting.
- **Included:**
 - Lubricants and related emissions
 - Electricity consumption (direct and indirect)
 - Machine depreciation (capital goods) and end-of-life (EoL)
 - Building infrastructure
- **Excluded:**
 - Fibre and yarn production (modelled upstream)
 - Textile/material waste (yarn losses and waste-yarn treatment), which is handled externally by the caller via the process-specific IOR
 - Sewing, dyeing, finishing and packaging

- Factory building beyond the amortised area share modelled in Section 4.5
- **Background data:** ecoinvent 3.12, system model: cut-off by classification.

Technical Overview

2.1 Process characteristics

Hosiery knitting uses small-diameter single-cylinder circular knitting machines designed for automatic toe closure and optional linking.

Production rates, power consumption and material losses are highly dependent on the gauge, feed configuration and automation level.

Key benchmarks:

- **Output:** ~300–400 pairs/day per standard machine; premium multi-feed models can exceed 800 pairs/day (~33 pairs/h).
- **Machine weight:** typically **230 - 320 kg**.
- **Energy:** **0.5 - 1.2 kWh/kg** (machine electricity only).
- **Lubricant consumption:** **0.05–0.10 L/kg**, depending on lubrication system efficiency.
- **Yarn waste:** **2–5%**, depending on toe-closing system and yarn type.

Inventory Data

Category	Flow	Amount	Unit	Comment
Product	Hosiery fabric (average mix)	1	kg	Representative of mixed cotton/synthetic yarn composition.
	Lubricant, average	0.008	kg	Lubricant: 8 g kg ⁻¹ fabric "Sandin et al.2019 pp45 - The lubricant was assumed to correspond to 8% of the weight of the knit."
	Air emissions from lubricant	0.008	kg	Equal to lubricant input for simplification.
Electricity / heat	Electricity mix	1.2	kWh	Average SEC derived from literature and manufacturer data: circular knitting ≈ 0.8–1.2 kWh/kg (Maity et al., 2021; Lonati GK series ≈ 0.5 kW for 0.9 kg/h output).
Indirect energy	Compressors, HVAC, lighting, robots	4.46	kWh	Indirect uses (air compressors ~6 bar, cooling, illumination, auxiliaries). Harmonised with other knitting processes for methodological consistency.
Capital goods	Building Machine (market for building machine)	0.0004	kg/kg fabric	Based on 300 kg machine amortised over 720,000 kg lifetime output (see below).
Waste to treatment	EoL Machine (Steel)	0.00035	kg	85% of total mass (steel) distributed per kg of fabric.
	Recycling (Other materials)	0.00006	kg	15% of total mass (motors, electronics, plastics).

4. Capital Goods and Building Infrastructure

Shared methodology: [Capital Goods & End-of-Life \(Machinery\) in LCA](#) | [Building Infrastructure in LCA](#). This section states only hosiery-specific inputs.

4.1 Equipment data

Machine	Weight (kg)	Source
Lonati GE616DF / GE516DF (family)	300	Lonati / Santoni technical brochures (2023)

4.2 Allocation parameters

Scenario	Years	Hours/day	Days/year	Productivity (kg/h)	Lifetime output (kg)	Allocation (kg/kg fabric)
Average	15	16	300	10	720,000	0.0004

4.3 End-of-Life (EoL) allocation

Material	Share (%)	Weight (kg)	Allocation (kg/kg fabric)
Steel	85	255	0.00035
Other materials	15	45	0.00006

These allocations show the portion of the machine infrastructure “consumed” per kilogram of hosiery fabric.

4.4 Representation in ecoinvent

Field	Entry
Dataset	`market for building machine {GLO} \
Amount	1.33E-06 unit/kg
Unit	unit
Product	building machine

Derivation: $300 \text{ kg machine} \div 720,000 \text{ kg lifetime output} = 0.000417 \text{ kg/kg}$; $\div 300 \text{ kg/unit}$ (Lonati machine mass as reference unit) = **1.33E-06 unit/kg**. Consistent with Brightway dataset.

4.5 Building Infrastructure

Shared methodology: [Building Infrastructure in LCA](#).

Process-specific value: $1.61\text{E-}05 \text{ m}^2/\text{kg}$ — derived from Lonati GE616DF3 envelope ($127.3 \text{ cm} \times 114.8 \text{ cm} \approx 1.46 \text{ m}^2$) with a higher cell factor for multi-machine layout and shared creel infrastructure. Comparable to seamless knitting (same value adopted). Physical plausibility cross-checked against multi-machine cell configurations.

ecoinvent activity: [market for building, hall {GLO} | cut-off](#) (unit: m^2) The building is represented by an ecoinvent-style construction dataset ("building, hall") whose reference flow is m^2 of industrial hall.

4.5.1 Allocation principle (area amortisation over lifetime output).

Building infrastructure is attributed to 1 kg of knitted hosiery by amortising the allocated hall area over the lifetime production associated with the knitting cell:

$$\text{Building intensity (m}^2\text{/kg)} = \frac{A_{\text{alloc}} \text{ (m}^2\text{)}}{Q_{\text{lifetime}} \text{ (kg)}}$$

$$Q_{\text{lifetime}} = \text{productivity (kg/h)} \times \text{hours/day} \times \text{days/year} \times \text{years}$$

where A_{alloc} is the hall area allocated to the knitting cell (machine footprint + access + yarn supply + maintenance space + internal logistics), and Q_{lifetime} is the total knitted output produced during the reference service life under the chosen operating regime.

4.5.2 Technology rationale.

Hosiery machines are physically small relative to other knitting technologies (individual machine envelope below 1.5 m²), but are typically deployed in multi-machine cells with shared yarn feed, collective creel systems, and common aisles. The resulting cell area, normalised by throughput, is comparable to seamless knitting, placing hosiery in the same intensity band. Additionally, the low mass per article and relatively slow cycle time per gram of output sustain a moderate building intensity per kg.

4.5.3 Practical implementation.

In the hosiery knitting foreground unit process, we add an input: *building, hall* (unit = m²) with amount = 1.61E-05 m²/kg.

4.5.4 End-of-life of the building.

The building dataset includes construction, maintenance and decommissioning within the background system model. No separate building EoL module is created in the foreground; only the amortised m²/kg input is provided.

4.5.5 Default value (this study). In the absence of site-specific A_{alloc} and Q_{lifetime} , the following reference intensity is applied, shared with the seamless technology group given comparable normalised throughput per allocated space:

■ Hosiery knitting: **1.61E-05 m²/kg**

Physical plausibility was cross-checked using machine envelope data from Lonati GE616DF3 (width 127.3 cm × depth 114.8 cm, envelope ≈ 1.46 m²) with a higher cell factor to account for multi-machine layout, shared yarn supply infrastructure, and aisles, yielding an allocated area consistent with the adopted default.

5. Indirect Energy

■ **Shared methodology:** [Indirect Energy in LCA — Shared Methodology \(Knitting\)](#).

Default value: 4.46 kWh/kg — ADEME & Ministère de la Transition Écologique (2024): $2.4 \div 0.35 \times 0.65 = 4.46$ kWh/kg. Harmonised across all knitting processes. **Already aggregated** with direct electricity in the Brightway exchange (total = 1.2 + 4.46 = 5.66 kWh/kg).

Subsystem distribution for reference:

Subsystem	Share (%)	Equivalent (kWh/kg)
Knitting machines	35	1.56
Compressors	25	1.12
HVAC	30	1.34
Lighting	6.5	0.29
Cleaning / auxiliaries	3.5	0.16

6. Waste Generation

Seamless hosiery knitting produces **minimal cutting waste**, in contrast to conventional cut-and-sew garments that lose 10–25% of material during trimming.

Industrial data (K&G Garment, 2025; GIZ 2023) show material utilisation above 95%, corresponding to **<5% total losses**.

In line with the knitting convention, the foreground dataset does **not** include an inline textile-waste exchange. Instead, the 2% default loss assumption is handled externally by the caller through the process-specific IOR, which scales upstream yarn input and routes the lost yarn to end-of-life treatment.

7. Key Assumptions Summary

Parameter	Average value	Comment
Electricity (machine)	1.2 kWh/kg	Lonati / literature average
Indirect energy (HVAC, compressors, etc.)	4.46 kWh/kg	Harmonised default for knitting
Lubricant	0.008 kg/kg	8 g/kg fabric typical maximum
Yarn loss rate (for IOR)	2%	Used only to derive the external IOR; no inline yarn-loss exchange is included in the dataset.
Machine mass allocation	0.0004 kg/kg	300 kg machine amortised
EoL (steel)	0.00035 kg/kg	85% share
Lost yarn treatment	External via IOR	Handled by the caller via the process-specific IOR and routed to waste treatment outside the dataset.

Input–Output Ratio (IOR)

Definition: $IOR = 1 / (1 - L)$, where L is the yarn loss rate (as a fraction). For every 1 kg of finished hosiery fabric, the upstream yarn input must be scaled by IOR.

How the loss rate is derived

Modern hosiery machines with automatic toe closure achieve defect rates below 2%. The 2% default is consistent with Accio (2023) industry benchmarks and Laursen et al. (2007) ranges. Yarn losses include thread breaks, machine setup waste, and separation thread consumption.

IOR value

Process	Loss (%)	IOR (kg/kg)	Input required (kg)	Loss per kg output (kg)	
Knitting, Hosiery \ Knitting Process	2%	1.0204	1.0204	0.0204	

How to apply: for 1 kg of hosiery fabric, multiply upstream yarn input by **1.0204**. The 0.0204 kg of lost yarn is routed by the caller to for example `market for waste yarn and waste textile {Row}`. No inline yarn-loss or textile-waste exchange is included in the dataset itself.

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